

Behavioral Simulation & Verification Report: Experiment 1

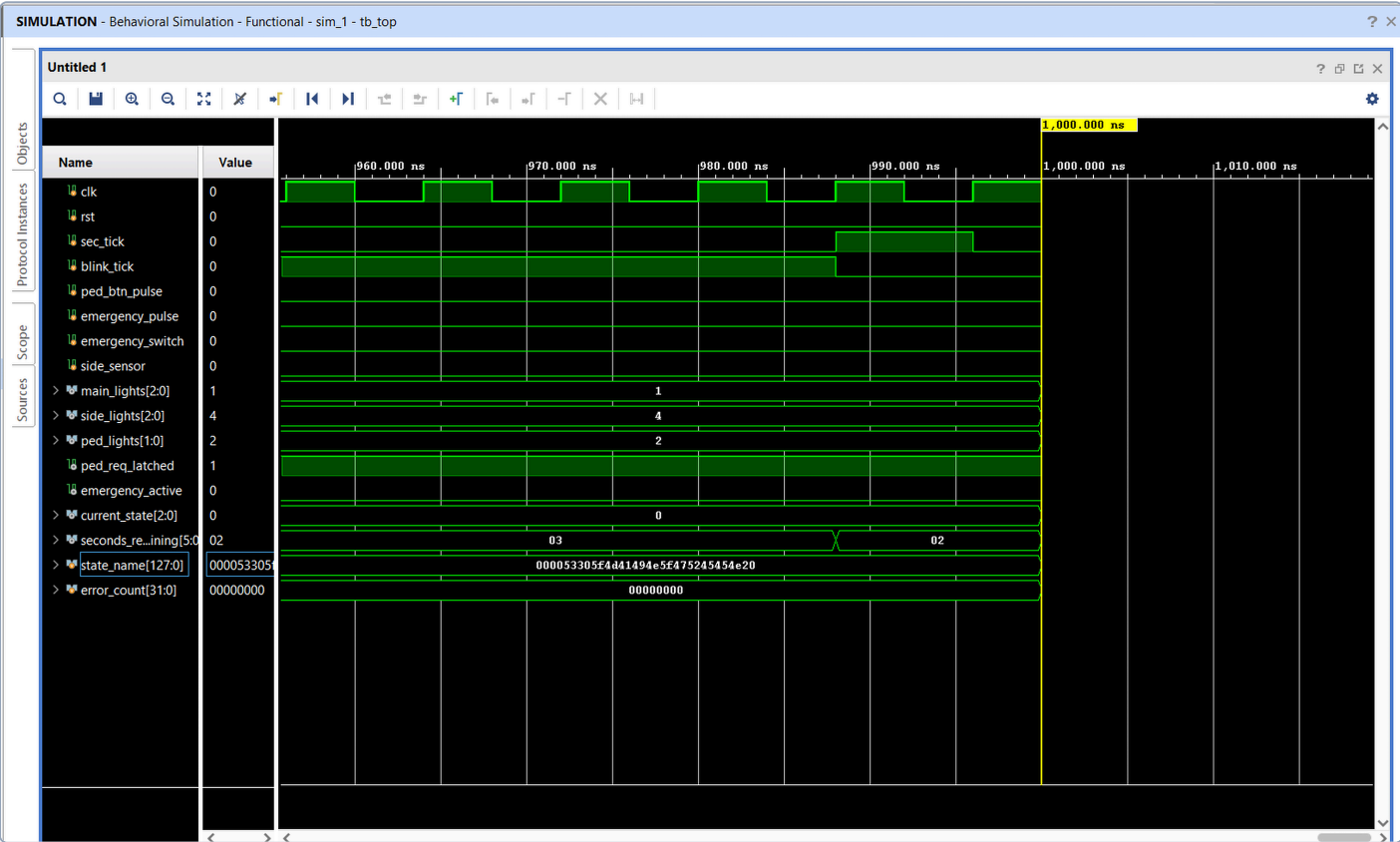
Design Under Test: traffic_fsm.v | Testbench: tb_top.v | Target: AMD PYNQ-Z2

1. Verification Matrix & Asserted Properties

Phase	Verification Objective	Monitored Signals	Result
Phase 1	Power-on Reset & Nominal Default State	current_state == S0_MAIN_GREEN, main_lights == 3'b001	PASSED
Phase 2	Normal Sequential Cyclic Transitions	S0 → S1 → S2 → S3 → S4 → S5 → S0	PASSED
Phase 3	Pedestrian Button Latching	ped_req_latched == 1, enters S6 after clearance, auto-clears	PASSED
Phase 4	Pedestrian Warning Flash	2 Hz toggle during final 2s of crossing cycle	PASSED
Phase 5	Emergency Vehicle Preemption	Mandatory safe 3s yellow clearance from Side Green → S7 Priority Green	PASSED
Phase 6	Hardware Safety Interlocks	Zero conflicting vehicular greens: !(main_green & side_green)	PASSED

Figure 1: Vivado Behavioral Simulation Waveform (tb_top.v)

1,000 ns Nanosecond-Scale Multi-Signal Behavioral Trace — Assertion Errors: 0



Waveform Trace Analysis: Multi-channel behavioral simulation trace in Vivado. The waveform demonstrates proper reset release, cycling across all 8 states, automated request latching on `ped_req_latched`, warning flasher modulation on `blink_tick`, and safe emergency corridor preemption with 0 assertion errors.

2. Console Execution Transcript

```
=====
STARTING TRAFFIC & PEDESTRIAN FSM CONTROLLER SIMULATION TESTBENCH
Platform: AMD PYNQ-Z2 (XC7Z020-1CLG400C) | Clock: 125 MHz
=====
[PASS] Initial State is S0_MAIN_GREEN (Main Green Active).
[PASS] Normal cyclic progression verified with proper clearance phases.
[PASS] Pedestrian request successfully latched (ped_req_latched = 1).
[PASS] Correctly entered S6_PED_CROSS (Pedestrian Walk Active).
[PASS] Pedestrian request latch automatically cleared after crossing.
[PASS] Verified: Controller DID NOT abruptly switch Side Green to Main Green. Safe 3s yellow clearance was enforced!
[PASS] Entered S7_EMERGENCY with Priority Corridor Main Green active.
[PASS] Assertion Check: (main_green & side_green == 0)
[PASS] Assertion Check: (ped_walk & (main_green | side_green) == 0)
>>>> ALL VERIFICATION TESTS PASSED SUCCESSFULLY! (0 ERRORS) <<<<
=====
```